

GEOGRAPHY

INDIAN GEOGRAPHY



INDIAN NATURAL VEGETATION

MODULE - 3

VEGETATION OF INDIA

- SOCIAL FORESTRY
- PROBLEMS OF INDIAN FORESTS

DIRECT USES OF FORESTS

- 1 Hard wood and softwood timber, fuelwood.
- 2 Grasses, bamboo and canes
- 3 Tans and dyes
- 4 Oils, gums and resins
- 5 Fibres and leaves
- 6 Drugs and spices
- 7 Edible products and animal products

INDIRECT USES OF FORESTS

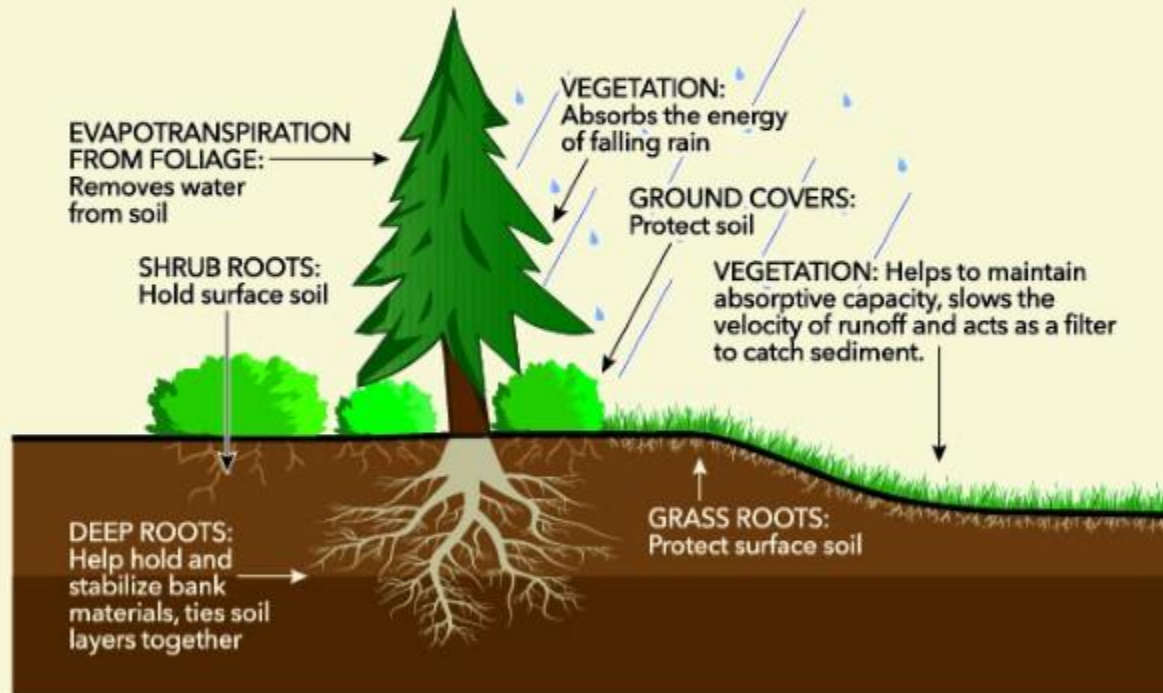


- 1** Apart from their direct uses described above, the forests are of immense use to man indirectly also.
- 2** They prevent soil erosion, regulate the flow of rivers and reduce the frequency and intensity of floods, check the spread of deserts, add to soil fertility and ameliorate the extremes of climate.

INDIRECT USES OF FORESTS

PREVENTION AND CONTROL OF SOIL EROSION

POINT NO:1



EFFECTS OF VEGETATION IN MINIMIZING EROSION

- 1 Forests play a significant role in the prevention and control of soil erosion by water and wind.
- 2 The destruction of forest cover leads to **increased runoff of rain water and its diminished seepage and storage in soil.**
- 3 The structure of the soil suffers, runoff increases and loosens the soil which is carried away to other regions.
- 4 The fertility of the soil is thus lost, and it becomes **barren and unproductive.**

INDIRECT USES OF FORESTS

PREVENTION AND CONTROL OF SOIL EROSION

POINT NO:1



5 The reckless destruction of forests in the Shiwalik hills, Western Ghats, Chhota Nagpur plateau and in the Deccan table-land has resulted in serious problem of soil erosion.

6 The most effective way to check soil erosion is to **stop reckless cutting of forests and to plant more trees.**

INDIRECT USES OF FORESTS

POINT NO: 2

FLOOD CONTROL

- 1 Roots of the trees absorb much of the rain water and use it slowly during the dry season. Thus they regulate the flow of water and help in controlling the floods.



- 2 Trees also act like millions of tiny dams and check the flow of water like a barrage. The forest cover thus acts as a rain-holder and a rain banker.
- 3 With the increased rate of deforestation, the frequency and the intensity of floods has increased in different parts of the Country.
- 4 In the absence of forests, the increased runoff along with sand and silt, especially after heavy downpour, comes in as a sudden rush and often gets blocked in the silted streams and causes devastating floods.

INDIRECT USES OF FORESTS

CHECKS ON SPREAD OF DESERTS

POINT NO: 3

- 1 Sand particles are blown away by strong winds in the deserts and are carried over long distances, thus resulting in the spread of deserts.



- 2 The roots of the trees and plants bind sand particles and do not permit their easy transportation by wind.
- 3 Therefore, the forests are a great instruments to put a check on the spread of deserts.
- 4 In the long run, the forests add humidity to the atmosphere and further help in checking the spread of deserts.
- 5 A great success has been achieved in checking the march of the Thar desert by planting a 650 km long and 8 km wide strip of trees on its periphery.

INDIRECT USES OF FORESTS

POINT NO:4

INCREASE OF SOIL FERTILITY

The fallen leaves of trees add humus to soil after their decomposition. Thus forests help in increasing the fertility of soil.



POINT NO:5

EFFECT ON CLIMATE

- 1 Forests have a far reaching effect on climate.
- 2 They **ameliorate the extremes of climate by reducing the heat in summer and cold in winter.**
- 3 They also influence the amount of rainfall through the process of transpiration.
- 4 They reduce the surface velocity of winds and retard the process of evaporation.

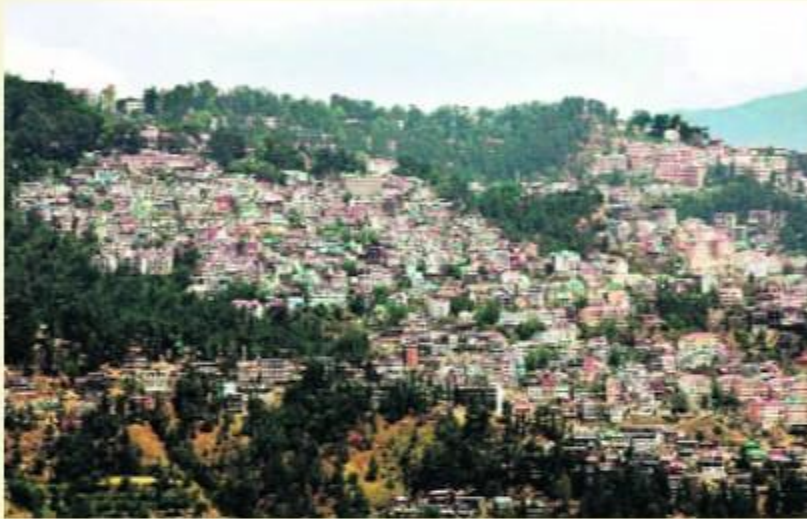
Indian forests face a number of problems which are both natural and manmade. Some of the major problems are briefly discussed as follows:

1 INADEQUATE AND DWINDLING FOREST COVER

- 1 The biggest problem of the Indian forests is the inadequate and fast dwindling forest cover.
- 2 It has already been mentioned that forests cover only around 21 per cent of the area against the required coverage of 33 per cent.
- 3 Even this small percentage of forest cover is seriously threatened by the **increasing demand for the major and minor products**.
- 4 These products are badly needed for fuel, building and to feed a large number of forest based industries.
- 5 Vast forest tracts have been cleared for **agriculture**.
- 6 **Shifting agriculture** in different parts of the Country has played havoc with forests.

PROBLEMS OF INDIAN FORESTRY

1 INADEQUATE AND DWINDLING FOREST COVER



7 India possesses a livestock population of over 412 million of which 270 million are bovine animals, about one-tenth of which graze in the forests.

8 Whenever forests are easily accessible, the livestock entirely depends on grazing in them.

9 A large part of our achievements made by virtue of afforestation are neutralised by diversion of forest land for non-forest use.

PROBLEMS OF INDIAN FORESTRY

2

LOW PRODUCTIVITY



1 Productivity of Indian Forests is very low as compared to some other Countries.

2 For example, annual productivity of Indian forest is only 0.5 cubic metre per hectare while it is 1.25 cubic metre per hectare in the USA, 1.8 cubic metre per hectare in Japan and 3.9 cubic metre per hectare in France.

PROBLEMS OF INDIAN FORESTRY

3

NATURE OF FORESTS AND THEIR UNECONOMICAL UTILISATION



- 1 The forests are **inaccessible**, **slow growing** and **lack in gregarious stands** in many parts of the Country.
- 2 Some of them are very thin and **comprise only of thorny bushes**.
- 3 These factors **make their utilization uneconomical** because there is a **good deal of wastage** and this makes it very expensive in spite of the cheap labour available in India.

PROBLEMS OF INDIAN FORESTRY

4

LACK OF TRANSPORT FACILITIES



- 1 One of the biggest problems faced by the Indian forests is the lack of proper transport facilities.
- 2 About 16 per cent of the forest land in India is inaccessible and does not have proper transport facilities.
- 3 It must be remembered that the major product of the forests is timber which is a cheap and bulky commodity.
- 4 As such it cannot afford high freights charged by the railways and roadways.

PROBLEMS OF INDIAN FORESTRY

4

LACK OF TRANSPORT FACILITIES



5 Therefore, Indian forests **cannot be economically exploited** without the availability of cheap and efficient transport facilities.

6 Unfortunately, in India, the **railways serve thickly populated areas only** and are not of much use of forests.

7 All weather roads in the forest areas **are badly lacking**. Water transport has only limited scope. Considering the facts we can easily say that transport with reference to forests is inadequate in India.

PROBLEMS OF INDIAN FORESTRY

5

FOREST FIRES



- 1 Large tracts of vegetal cover are destroyed every year by forest fires.
- 2 Forest fires in India are most destructive in dry season.
- 2 Insufficiency of properly trained personnel is a big handicap.

PROBLEMS OF INDIAN FORESTRY

6

PLANT DISEASES, INSECTS AND PESTS



Sal borer and its effect on Sal tree

1 Large tracts of forest cover suffer from plant diseases, insects and pests which leads to considerable loss of forest wealth.

2 For example, thousands of hectares of sal forests in Madhya Pradesh and Chhattisgarh are being threatened by sal borer for which no remedial measures have been adopted so far.

3 Forest officials are only using the primitive methods of hiring the tribal to catch and kill the insect.

PROBLEMS OF INDIAN FORESTRY

7

OBSOLETE METHODS OF LUMBERING



- 1 In most of the Indian forests, obsolete methods of lumbering are practiced.
- 2 This system leads to a lot of wastage and low productivity.
- 3 Large quantities of inferior wood which could be put to better use through seasoning and preservation treatment remain unutilized or go waste.
- 4 Saw mills use old obsolete machinery and do not get proper power supply.

8 LACK OF COMMERCIAL FORESTS

- 1 In India most of the forests are meant for **protective purposes** and **commercial forests** are badly lacking.
- 2 Growing awareness about environmental degradation has forced us to look at forest wealth as a protective agent for environment rather than treating all forests as a purely commercial commodity.

9 LACK OF SCIENTIFIC TECHNIQUES

- 1 Scientific techniques of growing forests are also lacking in India.
- 2 Only natural growth of forests takes place in India whereas in many developed countries new scientific techniques are being used through which tree growth is quickened.
- 3 A large number of trees are malformed or consist of species which are slow growing and poor yielders.

PROBLEMS OF INDIAN FORESTRY

10

UNDUE CONCESSIONS TO TRIBALS AND LOCAL PEOPLE



- 1 In vast forest tracts, tribals and local people have been granted customary rights and concessions for free-grazing as well as removing timber fuel and minor forest products.
- 2 They are also allowed to continue with age-old shifting cultivation.
- 3 These practices have led to the reduction in forest yield. In addition, there has been encroachment on these forests by the village people inhabiting the peripheral areas.

PROBLEMS OF INDIAN FORESTRY

REMEDIES

At present there is an urgent need of undertaking silvicultural operations on a large scale. This can be done through the following measures:

- 1** Intensive development schemes for afforestation should be adopted. High yielding varieties should be planted in suitable areas.
- 2** Improved techniques of logging and extraction should be used.
- 3** Proper transport facilities should be provided to remote and inaccessible forest areas.
- 4** Saw mills should get uninterrupted power supply. Latest techniques of seasoning and preservation are necessary to avoid wastage.
- 5** Proper arrangements to save forests from fires and plant diseases can go a long way to solve several problems.
- 6** A thorough inventory of forest resources is necessary to make an accurate assessment of our forest resources and make plans for their proper use.
- 7** Shifting cultivation should be discouraged and tribals depending on this type of cultivation should be provided with alternate sources of livelihood.
- 8** People associated with forest protection should be properly trained.

SOCIAL FORESTRY



- 1** Social forestry means the **management and protection of the forests as well as afforestation of barren lands** aimed at helping in environmental, social and rural development as against the traditional objective of securing revenue.
- 2** It is aptly described as forestry of the people, by the people and for the people.
- 3** The main thrust of social forestry is to **reduce pressure on the traditional forests** areas by developing plantations of fuelwood, fodder and grasses.

SOCIAL FORESTRY

4 The term social forestry was used for the first time by the **National Commission on Agriculture in 1976**, to denote tree raising programmes to supply firewood, small timber and minor forest produce to rural population.

5 This commission divided social forestry into

- i. Agro forestry
- ii. Expansion forestry
- iii. Rehabilitation of degraded and low grade forests and
- iv. Recreation forestry.

6 The programme was formally launched in 1978 and it became an integral part of the Sixth Plan in 1980.

7 In the second Forestry Conference (1980) it was decided that social forestry scheme should be given priority over barren and waste land, community land and lands along the roads, railways and canals.

MAIN COMPONENTS OF SOCIAL FORESTRY

Agro forestry, community forestry, commercial farm forestry, non-commercial forestry and urban forestry are the main components of social forestry.

1

AGRO FORESTRY



1

Agro forestry involves the raising of trees and agricultural crops either on the same land or in close association in such a way that all land including the waste patches is put to good use.

2

This enables the farmer to get food, fodder, fuel, fruit and timber from his land.

3

The land gives maximum production and provides employment to rural masses.

2

COMMUNITY FORESTRY



1 Community forestry involves the raising of trees on public or community lands aimed at providing benefit to the community as a whole.

2 Although the plants and seedlings are provided by the forest department, the protection of plantations is primarily the responsibility of the community as a whole. India is second only to China in community forestry.

3 Remarkable success has been achieved in community forestry in some major States like Gujarat, Tamil Nadu, Rajasthan, Kerala, Karnataka, Uttar Pradesh, Haryana, Orissa and Himachal Pradesh.

3 COMMERCIAL FARM FORESTRY

- 1 Commercial Farm Forestry involves growing of trees in the fields on place of food and other agricultural crops.



- 2 In this way, the farmers grow trees for direct commercial gain. This type of forestry has become very popular in Haryana, Punjab, Gujarat, Uttar Pradesh, Rajasthan and in some parts of south India.
- 3 Farmers in these areas grow eucalyptus, poplars and casuarina in their land and sell them to paper, pulp, rayon and match industries and also use them for making packing cases and fuel.
- 4 The State Governments are encouraging this forestry in a big way.

4 NON-COMMERCIAL FARM FORESTRY



- 1 Non-Commercial Forestry involves tree planting by farmers on their own land for their own use and not to sale.
- 2 The land mainly used for this purpose includes margins of agricultural fields, wastelands and marginal lands which are not usually cultivated, grasslands and pastures and land around homes and cowsheds.
- 3 This type of forestry is practiced in Gujarat, Haryana, Kerala and Karnataka for obtaining fuel, fodder and fruit.

5

URBAN FORESTRY



- 1 Urban Forestry pertains to raising and management of trees on **public** and **privately owned** lands in and around urban centres.
- 2 It includes **green belts**, **roadside avenues**, **recreational parks**, **wildlife parks** etc.
- 3 Its main objectives are reduction of **environmental pollution**, **recreation** and **improving aesthetic values**.

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